TikTok Claims Classification Project

Logistic Regression Model — Executive Summary

■ ISSUE / PROBLEM

TikTok's Operations Lead requested an analysis to understand how different video characteristics are associated with whether a user is verified. The team previously observed that verified users are more likely to post opinions. This logistic regression model explores that relationship and helps inform the final claims classification model.

■ RESPONSE

A logistic regression model was built using `verified_status` as the binary outcome variable. Logistic regression was selected over linear regression because the outcome is categorical. The model was evaluated using a confusion matrix and classification report.

Model specifications:

- Outcome: verified_status (binary)
- 7 input features selected
- video_like_count excluded — high multicollinearity with view count
- Class imbalance corrected via upsampling
- 75/25 train/test split, random_state=0

■ MODEL VARIABLES

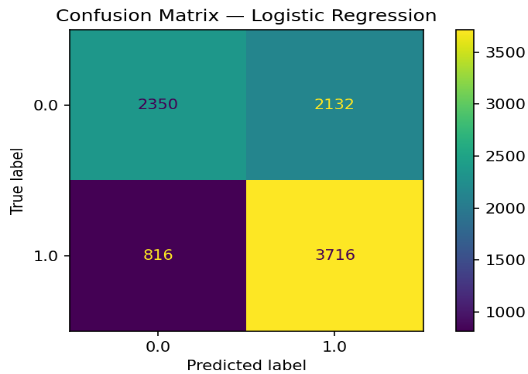
Feature	Type	Included?	Reason
video_duration_sec	Numeric	■ Yes	Key video characteristic
claim_status	Categorical	■ Yes (encoded)	Strong predictor of verified status
author_ban_status	Categorical	■ Yes (encoded)	Account behavior signal
video_view_count	Numeric	■ Yes	Engagement metric
video_share_count	Numeric	■ Yes	Engagement metric
video_download_count	Numeric	■ Yes	Engagement metric
video_comment_count	Numeric	■ Yes (capped)	Engagement metric (outliers capped)
video_like_count	Numeric	■ Excluded	Highly correlated with view_count (multicollinearity)
#, video_id	Identifier	■ Excluded	Not predictive

■ RESULTS

Metric	Verified	Not V
Precision	74%	64%
Recall	52%	82%
F1-score	61%	72%
Accuracy	—	—

The model achieved 67% overall accuracy. Recall for the 'not verified' class is 82% — meaning the model correctly identifies 82% of unverified accounts. This is the more important metric for this use case.

Confusion Matrix



TN=2,350 · FP=2,132 · FN=816 · TP=3,716

■ KEY INSIGHTS

The logistic regression model reveals several important insights for the claims classification project. The two key findings from this analysis were:

Claim status is the strongest predictor

The model coefficient for `claim_status_opinion` is +1.85, by far the largest coefficient. Opinion videos are strongly associated with verified accounts. This confirms the earlier hypothesis and validates claim status as a critical feature for the final model.

Model is acceptable but not excellent

At 67% accuracy and 82% recall, this logistic regression model provides a useful baseline. More sophisticated models (Random Forest, XGBoost) with NLP features from video transcription text are recommended as next steps for improved performance.

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